MARTIAN MOONS EXPLORATION

MMX

Earth to Phobos. Samples home to Earth.

Investigating the origin of Mars's moons, from orbital surveys to samples in Earth laboratories.

MISSION PROFILE

2026—2031

PRE-LAUNCH PLAN

2026 — 19 Oct UTC · H3 F10 launch

2027 — Arrive at Mars

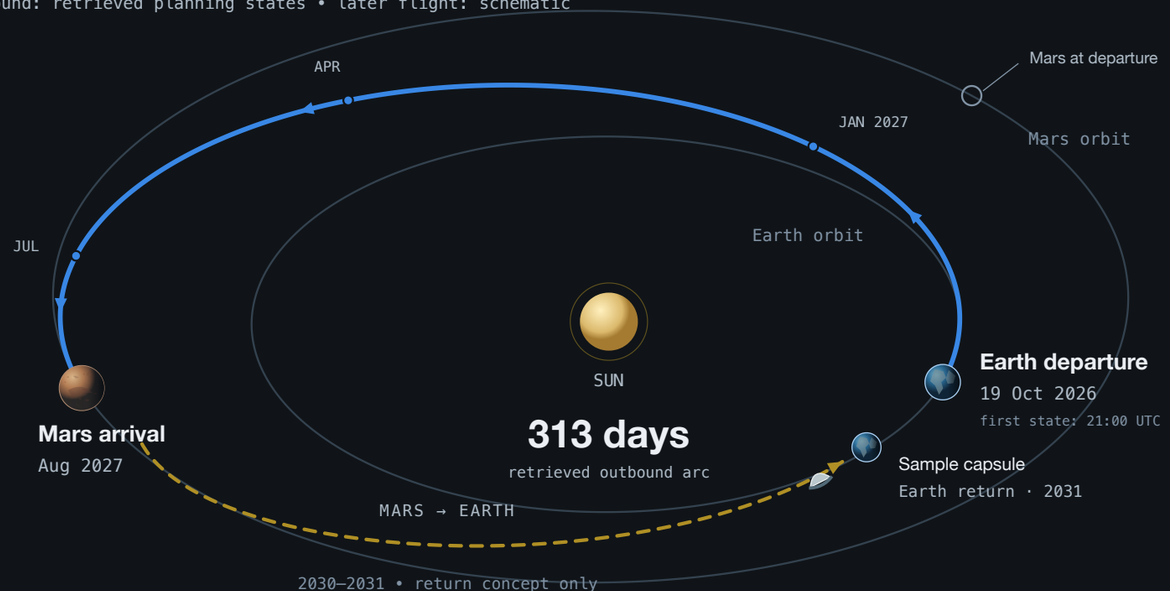
2027–2030 — Explore & sample Phobos

2030 — Leave Mars

2031 — Return samples to Earth

01 The interplanetary journey

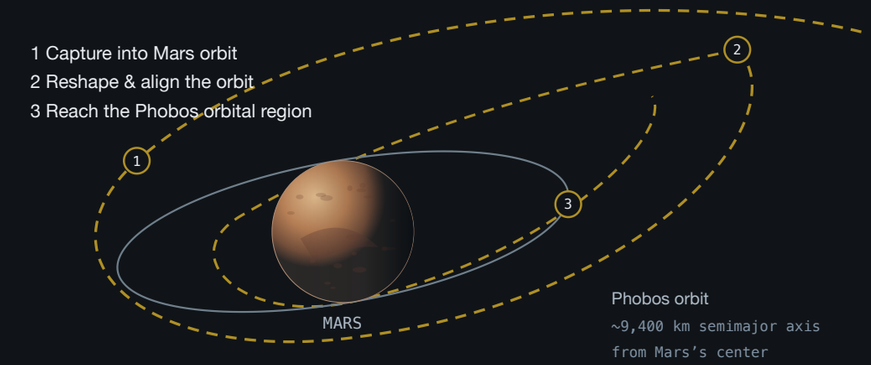
Outbound: retrieved planning states • later flight: schematic

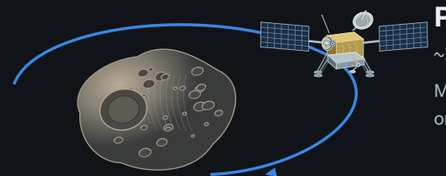


Projected J2000 ecliptic plane • planet/body sizes enlarged

02 Capture, then fly with Phobos

Mars-centered concept • not a reconstructed insertion sequence





PHOBOS

~27 × 22 × 18 km

MMX flies nearby in quasi-satellite orbits while both travel around Mars.

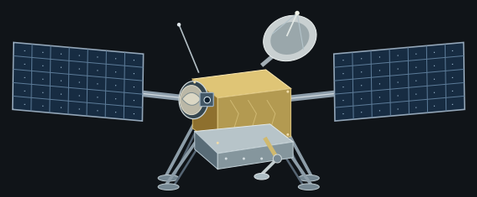
Relative-motion sketch near Phobos

Observe → descend → land → collect → ascend

The Martian moon is the sample target—not Mars itself.

03 The probe

Illustrated flight hardware • not to scale



At Phobos: exploration + return modules together.

Propulsion module discarded after Mars capture.

04 Two ways to collect Phobos

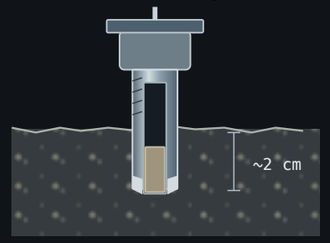
Complementary sampling of the surface and subsurface

≥10 g

Total return target • two sites planned

C-SMP / CORING

Drive a hollow corer into regolith



A tube penetrates below the surface, retaining a core of regolith for return.

JAXA / subsurface regolith

P-SMP / PNEUMATIC

Nitrogen jets at a landing foot



Nitrogen lifts surface grains. Further gas flow carries them up the tube and into a sample canister.

NASA · Honeybee Robotics / surface grains

Data: JAXA / ISAS • JPL Horizons -239 • DE440s | Geometry beyond the outbound arc is schematic.

Independent mission study | Illustration: Mickey